



ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

Department of Pharmacy and Biotechnology - FaBiT

First cycle degree in Genomics

Spatial Transcriptional Entropy reveals Context Dependent Transcriptional Programs and Spatial Heterogeneity in Glioblastoma

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Academic Year 2025/2026

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Acknowledgments

I would like to thank my supervisor, Professor Fabrizio Ferrè, for his guidance and support and valuable feedback throughout this project.

I would like to thank my mother for being my support and for encouraging me through every setback. I would also like to thank my father, who I believe watches over me from above, for giving me the strength to face every challenge.

I would also like to thank my friends from the Genomics course for their support, both academically and personally, and for standing with me throughout the course until the end.

Finally, I would like to thank myself for staying committed until the end and for continuing to push forward even in moments when I felt like giving up. This journey has been challenging but also highly rewarding, and I am grateful for everyone who supported me along the way.

Abstract

Glioblastoma (GBM) is a highly aggressive brain tumor known for strong molecular and spatial heterogeneity. This study analyzed spatial transcriptomic data from multiple GBM samples to investigate transcriptional heterogeneity using transcriptional entropy combined with functional enrichment and known GBM transcriptional signatures. Transcriptional entropy was used to quantify heterogeneity across spatial spots and samples were further analyzed using mean entropy, standard deviation of entropy and Gaussian Mixture Modeling (GMM). Functional enrichment analysis was performed on spatially defined entropy regions and results were compared with GBM subtype signatures (Glioma stem cells (GSC), mesenchymal, proneural, classical and neural).

Six spatial transcriptomic samples generated using 10X Genomics Visium platform revealed clear differences in the relationship between transcriptional entropy and biological programs across the sample. Some tumors such as GBM4, showed strong overlap between high entropy regions and GSC signatures, especially at proneural mesenchymal transition (PMT) regions. In contrast, GBM2 and GBM5 showed entropy patterns largely independent of stemness and instead linked to immune or segregated transcriptional programs. GBM1, GBM3 and GBM5 showed weak or negative association with stemness signature suggesting the entropy does not always reflect stem-like behaviours. GBM6 identified as peritumoral tissue showed entropy patterns driven mainly by neural and normal brain related processes.

Overall, ribosomal and translational activity was consistently enriched across all samples, indicating shared transcriptional programs in both tumor and peritumoral tissue. However, other processes such as immune response, metabolism and developmental signaling varied across samples. These findings show that transcriptional entropy is highly context-dependent and reflects different biological processes depending on tumor state and microenvironment. This highlights the complexity of GBM and supports the idea that multiple biological programs contribute to tumor heterogeneity.

1. Introduction

1.1 Glioblastoma

GBM is one of the most common and aggressive primary malignant brain tumors in adults and is classified as a grade IV astrocytoma by the World Health Organization [1]. GBM has poor prognosis with a median survival of approximately 14-18 months after diagnosis and five year survival rate that doesn't exceed 5%, despite advances in imaging and molecular diagnostics and introduction of integrated treatment strategies [2]. Current standard treatments consists of maximal surgical resection, radiotherapy and temozolomide (TMZ) chemotherapy however, several obstacles limit the effectiveness of these treatments [3, 4]. Complete surgical resection is difficult due to the highly infiltrative growth pattern of GBM, as removal of infiltrating tumor cells may result in neurological damage. Residual tumor cells at the invasive margins frequently contribute to rapid tumor recurrence [1, 5]. GBM also exhibits resistance to radiotherapy and TMZ chemotherapy through O⁶-methylguanine-DNA methyltransferase (MGMT) mediated DNA repair [1]. The tumor microenvironment (TME) further promotes tumor survival and therapy resistance through angiogenesis, metabolic adaptations, activation of tumor promoting transcriptional programs and suppression of anti-tumor immune response, highlighting the need for more effective therapeutic strategies [1].

Transcriptional profiling studies have classified GBM into mesenchymal, proneural, classical and neural subtypes, based on characteristic molecular alterations in *PDGFRA*, *IDH1*, *EGFR* and *NF1* [6]. However, subsequent studies suggest that neural subtype may partially reflect contamination of normal brain tissue than a true tumor subtype [7]. GBM exhibit spatial niches including perivascular, hypoxic and invasive niches, which create specialized microenvironments that contribute to tumor heterogeneity and disease progression [8]. In addition, GBM contains Cancer Stem Cells, known as Glioma Stem Cells (GSC) which exhibit self renewal capacity and ability to differentiate into multiple neural lineages [1]. Identification of GSC associated regions is important to understand GBM biology and develop targeted therapies.

Single-cell RNA sequencing (scRNA-seq) has provided insights to understand the cellular architecture and heterogeneity of GBM. However, it fails to capture the spatial organization of the tumor due to tissue dissociation during sample preparation, leading to loss of spatial information and cellular organization within the tumor [2, 9]. Therefore, approaches that are capable of preserving the spatial information are needed to better understand the biological organization and transcriptional heterogeneity of GBM.

1.2 Spatial Transcriptomics

Spatial heterogeneity in GBM is driven by diverse cell populations, distinct tumor niches and interactions between other GBM and TME components, making it difficult to analyse using conventional transcriptomic approaches [10]. Bulk RNA sequencing measures the average gene expression across a large population of cells, resulting in the loss of information individual cellular states and spatially distinct transcriptional programs [11]. To overcome this limitation, single cell RNA sequencing (scRNA-seq) emerged, allowing researchers to uncover cellular heterogeneity, identify rare cell types, and track developmental trajectories [11]. However, scRNA-seq requires dissociation of viable cells, which could disrupt the tissue architecture and spatial relationships between cells, potentially inducing cellular stress and cell death during sample preparation [9]. Therefore, Spatial Transcriptomics emerged, allowing researchers to map gene expression while preserving tissue structure and spatial context, which could not be achieved before using bulk RNA sequencing and scRNA-seq [11].

In GBM, spatial transcriptomics allows the identification of distinct tumor regions and their associated molecular pathways, spatially organized transcriptional programs and cellular interactions in the TME by integrating gene expression data with histological information [9, 12]. Despite its ability to preserve spatial information and reveal region specific transcriptional programs, quantification of tissue heterogeneity remains challenging. Therefore, entropy-based metrics have emerged as powerful approaches to quantify transcriptional heterogeneity and measure cellular plasticity, differentiation potential and transcriptional disorder [13].

1.3 Transcriptional Entropy

Transcriptional entropy is a measure of disorder, uncertainty or variability in gene expression states. In biological systems, Shannon entropy is the most widely used entropy, where it quantifies the uncertainty or variability by measuring the distribution of gene expression states across different cell states [14]. Generally, higher entropy values reflect more heterogeneous and less constrained gene expression patterns, while lower entropy values are associated with more stable and specialized transcriptional programs [15].

In scRNA-seq studies, entropy has been widely applied to quantify intercellular transcriptomic heterogeneity, cellular plasticity and differentiation potential [16]. Additionally, some scRNA-seq studies have used entropy as a measure of variability and stemness [17]. However, this study suggests that the biological interpretation of entropy is context dependent, as similar entropy levels may reflect different underlying biological processes based on tissue type and cellular composition.

In Spatial Transcriptomics, entropy based approaches can be used to quantify transcriptional heterogeneity while preserving spatial information within the tissue architecture. In complex tissues such as GBM, where cellular states are not uniformly distributed but organized into

spatially distinct niches, entropy provides a quantitative measure to capture spatial variability in gene expression and identify regions of increased transcriptional disorder [15, 18]. Therefore, transcriptional entropy represents a quantitative framework for assessing spatial heterogeneity and cellular plasticity when integrated with spatial and biological context. In this study, entropy is applied to ST data to investigate the organization of transcriptional heterogeneity in GBM.

1.4 Purpose of the study

GBM is highly heterogeneous due to the presence of diverse cellular populations, spatial niches and cellular plasticity. This spatial diversity drives different transcriptional programs and investigating these programs are important for understanding tumor biology and developing targeted therapies [19]. Although scRNA-seq enables transcriptomic profiling at cellular level, it loses spatial information during tissue dissociation. Therefore, spatial transcriptomics as an approach to overcome this limitation by preserving tissue architecture while measuring gene expression [9].

Despite its ability to retain spatial information, spatial transcriptomics still lacks a robust framework to characterize spatial heterogeneity across spatial regions. Previous entropy based analyses in scRNA-seq studies have successfully quantified cellular heterogeneity and transcriptional variability [17, 20, 21]. Using the same concept, this study applies transcriptional entropy as a quantitative measure to investigate spatial heterogeneity in GBM.

For effective profiling of transcriptional variability across spatial regions, both mean entropy and standard deviation of entropy were incorporated into the analysis. Mean entropy was used to quantify the overall transcriptional disorder within each spot, while standard deviation was used to capture gene specific entropy patterns across the tissue. These measurements enabled the identification of spatial regions with distinct entropy profiles and allowed the investigation of whether transcriptional heterogeneity is spatially organized within the GBM tissue architecture. In addition, these spatial regions were further characterized through functional enrichment analysis and comparison with GBM associated transcriptional programs including GSC and GBM subtype signatures. Therefore, this study aims to investigate how transcriptional entropy reflects spatial heterogeneity in GBM within different cellular and biological contexts. In particular, the study aims to determine whether regions with elevated entropy consistently correspond to stemness related transcriptional programs or whether they reflect distinct biological processes depending on tissue context.

2. Materials and Methods

2.1 Dataset and preprocessing

Spatial transcriptomics data from six patients who required surgical resection were obtained from Gene Expression Omnibus (GEO) under the accession number GSE194329. The dataset consisted of three IDH-wildtype primary GBM samples (GBM^{IDHwt}), two IDH-mutant secondary GBM samples (GBM^{IDHmut}) and one peritumor sample. The data were originally generated by Yanming Ren et al. using the 10x Genomics Visium platform and are described in detail in the original publication [22].

The samples were originally labeled as GBM1, GBM2, GBM3, GBM4, GBM5_1, GBM5_2. For simplicity, the last two samples were renamed as GBM5 and GBM6 respectively. GBM2, GBM3, GBM5 corresponded to GBM^{IDHwt}, whereas GBM1 and GBM4 represented GBM^{IDHmut}. GBM6 represented the peritumoral tissue and was used as a reference for comparison with tumor associated entropy profiles. Peritumoral tissue represents an intermediate interface between the tumoral and the healthy tissue region and reflects biological interactions occurring between normal brain tissue and the TME [23, 24].

All samples contained gene expression profiles for 36,601 genes with a varying number of spots per sample: 4,764 spots for GBM1, 2,891 for GBM2, 1,775 for GBM3, 2,540 for GBM4, 2,203 for GBM5 and 2,171 for GBM6. Each sample was normalized individually using the SCTransform function implemented in the Seurat package (version 5.4.0) in R [25]. SCTransform performs variance stabilizing transformation on unique molecular identifier (UMI) counts by fitting a regularized negative binomial model for each gene. UMIs are short molecular barcodes attached to RNA molecules during library preparation that allow accurate quantification of transcripts by distinguishing original molecules from PCR duplicates. The resulting normalized gene expression matrices were used for downstream analyses.

All analyses were performed in R (version 4.4.1). The Seurat package [25] was used for spatial transcriptomic data handling and visualization, mclust [26] for classification using Gaussian Mixture Model (GMM), clusterProfiler [27] and DAVID [28] for Gene Ontology and KEGG pathway enrichment analyses, matrixStats [29] for row-wise and column-wise standard deviation calculation, ggplot2 [30] for data visualization and pheatmap [31] for heatmap visualization.

2.2 Entropy calculation

Shannon entropy was used to calculate the transcriptional entropy as it is the most widely used measure of entropy in biological systems and has been used to assess cellular heterogeneity and stemness in scRNA-seq studies [14, 17]. In single cell transcriptomics, entropy can be defined

at multiple levels, including IntRACellular entropy which captures variability within a cell and IntERCerllular entropy which captures variability across cells [17]. In this study, IntERcellular entropy concept was adapted by treating each spatial spot as an independent transcriptional unit, enabling the quantification of spatial transcriptional heterogeneity across the tissue. The gene-wise contribution to Shannon entropy was calculated for each gene within each spatial spot using the following equation:

$$\hat{H}_{i,j} = -\hat{p}_{i,j} \log_2(\hat{p}_{i,j}) \quad (2.1)$$

Here, $\hat{p}_{i,j}$ represents the normalized frequency of expression of gene i in spot j , calculated as the expression of gene i divided by the total expression of all genes within that spot. $\hat{H}_{i,j}$ represents the transcriptional entropy contribution of gene i in spot j .

Unlike the standard Shannon entropy equation which sums entropy contribution into a single value per spot, this study retained the individual entropy contribution of each gene without summation, preserving gene-specific spatial variability across the tissue regions. This resulted in a gene x spot matrix where each entry represented the contribution of gene i to the overall transcriptional entropy of spot j . These matrices allows subsequent downstream analyses to quantify both the mean and standard deviation of the entropy per spot and variability of spatial regions, as described in the next section.

Entropy values were rescaled using min-max scaling where the values were scaled between zero and one, to facilitate cross sample comparison using the following equation.

$$X_{i,j,scaled} = \frac{X_{i,j} - X_{min}}{X_{max} - X_{min}} \quad (2.2)$$

where $X_{i,j}$ is the entropy value of the gene i in the spot j , while X_{min} and X_{max} represent the global minimum and maximum entropy values across all the samples, respectively.

Mean entropy per spot was computed using the scaled entropy matrix using colMeans function in R. In addition, standard deviation of entropy was computed both across genes within each spot using colSds function and across spots for each gene using rowSds function from matrixStats package [29] in R. This allowed the characterization of both spatial heterogeneity across tissue regions and the variability of genes associated with the entropy across the samples.

2.3 Classification of Discrete Spatial Entropy Regions

Gaussian Mixture Model (GMM) was used to classify discrete spatial entropy regions by incorporating both mean and standard deviation of scaled entropy, to capture average transcriptional disordered with entropy heterogeneity. GMM uses a probabilistic model that models data as a combination of multiple Gaussian distributions, each characterized by its own mean and variance, weighted by a mixing coefficient. It uses Expectation-Maximization (EM) algorithm [32]

to iteratively estimate model parameters, making the approach robust for identifying heterogeneous transcriptional regions in complex transcriptomics datasets [33]. The same framework was applied in this study, to identify discrete spatial entropy regions.

The classification was performed using Mclust function from the mclust package (version 6.1.2) in R [26]. Clustering was based on a combined score of mean entropy and standard deviation per spot. The optimal number of components for each class was determined using the Bayesian Information Criterion (BIC), which balanced model fit and model complexity by penalizing excessive parameters and reducing the risk of overfitting [34]. Each spot was assigned to the component with the highest posterior probability and the resulting classifications were stored in the metadata of the Seurat object under the name "group".

Prior to GMM classification, spots with extreme combined mean and standard deviation entropy scores were identified independently within each sample, using the interquartile range (IQR) using the following equations.

$$\text{Lower bound Outliers} = Q_1 - 1.5 \times (Q_3 - Q_1) \quad (2.3)$$

$$\text{Upper bound Outliers} = Q_3 + 1.5 \times (Q_3 - Q_1) \quad (2.4)$$

where Q_1 represents the 25th percentile, Q_3 represents the 75th percentile and $Q_3 - Q_1$ represents the IQR corresponding to the middle 50% of the distribution. Spots identified outside these thresholds were classified as outliers. These spots represented spots with extremely high entropy profiles relative to the distribution within each sample and were assigned a separate class label called "outlier". To preserve their biological distinctness, these spots were excluded from GMM classification but retained in all downstream analyses.

2.4 Gene and Pathway Enrichment

Gene Ontology (GO) enrichment and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway enrichment were performed using clusterProfiler package in R [27]. For each discrete spatial region, top 100 genes with highest mean entropy across spots belonging to that region were selected for enrichment analysis. Genes were compared against OrgDb annotation database for Homo sapiens, with a statistical significance set to $p < 0.05$ and pAdjustMethod set to Benjamini Hochberg multiple testing correction method. GO enrichment analysis was performed for Biological Process (BP) ontology, while KEGG pathway enrichment was used to identify enriched signalling and metabolic pathways. The background gene universe was defined as all genes detected in the gene x spot entropy matrix to avoid bias associated with using the complete human genome as background.

Before shared and unique gene enrichment analysis, genes with high entropy variability were visualized using a heatmap generated from gene level standard deviation values across the samples using the R packages pheatmap [31]. Only the top 15 genes with high entropy

variability were selected for a clear visualization. Genes exhibiting similar variability patterns were grouped into clusters based on hierarchical clustering of the heatmap. Genes in these clusters were enriched using DAVID [28] functional annotation tool. This analysis enabled characterization of biological themes related to distinct entropy variability patterns observed across glioblastoma and peritumor samples.

For the standard deviation gene analysis, genes with high entropy variability were identified independently for each sample using a 90th percentile threshold, which was applied to the gene x spot standard deviation of entropy matrix. Genes that shared between the tumor samples (GBM1-GBM5), genes shared between the tumor samples and the peritumor sample (GBM6) and genes that are unique to each sample were identified beyond that threshold. GO enrichment analysis with similar parameters used for the previous enrichment analysis were used. However, in this case all GO categories (biological processes, molecular functions and cellular components) were included instead of analyzing biological process only. This was done because using only biological process ontology resulted in incomplete or limited enrichment in several samples, while all GO categories improved the functional coverage across most samples. KEGG pathway enrichment was performed to identify metabolic pathways related to those genes.

During pathway enrichment, about 5-13% of genes were lost when gene symbols had to be converted to entrez ids. This happens when some genes like mitochondrial genes, lncRNA genes, pseudogenes do not contain entrez ids. The results were plotted and visualized on a dotplot using the library `enrichplot` [35]. `compareCluster` function from `clusterProfiler` [27] package was used for comparative enrichment analysis between multiple gene lists.

2.5 Module Score Validation

Module score validation was performed using `AddModuleScore` function from the `Seurat` package [25] in R. It calculates the average expression levels of predefined gene programs at spot level while subtracting the aggregated expression values from randomly selected control gene sets. Genes are grouped into expression bins according to their average expression levels, where control genes are randomly selected from each bin to reduce expression level bias afterwards. Although `AddModuleScore` is commonly applied for scRNA-seq analysis, it is compatible with any `Seurat` object, including spatial transcriptomics datasets which is consistent with previous applications [36]. Module scores were calculated using gene expression values stored in SCT assay of the `Seurat` object.

Module scores were computed for GSC markers (*CD133*, *CD15*, *CD44*, *CD70*, *S100A4*, *CD49f*, *L1CAM*, *A2B5*, *SOX2*, *NANOG*, *OLIG2*, *MYC*, *MSI1*, *NES*, *OCT4*, *ALDH1A3*) [37, 38]. In addition, GBM subtype associated marker sets were analysed, including mesenchymal markers (*YKL40*, *CD44*, *STAT3*, *CHI3L1*, *NF1*, *PTEN*, *MET*, *VIM*) [6, 7], proneural markers (*PDGFRA*, *OLIG2*, *NKX2-2*, *DCX*, *SOX*, *ASCL1*, *DLL3*, *TCF4*) [6, 7], classical markers (*EGFR*, *NES*, *NOTCH3*, *JAG1*, *LFNG*, *SMO*, *GAS1*, *GLI2*) [6, 7] and neural markers (*NEFL*,

SYT1, GABRA1, SLC12A5) [6, 7].

2.6 Fisher's exact test

Fisher's exact test [39] was performed to determine whether the overlap between transcriptional entropy regions and GSC regions occurred by chance. Fisher's exact test was selected over Chi-square test because it directly evaluates the association between categorical variables using exact probabilities derived from the hypergeometric distribution and is appropriate when cell frequencies are small [40]. For each sample, spots were classified as high mean entropy spots if their scaled mean entropy exceeded the 75th percentile of the entropy distribution. Similarly, spots were classified as high GSC spots if their module score exceeded the 75th percentile of the GSC score distribution. A 2x2 contingency table was constructed having high entropy versus low entropy spots and high GSC versus low GSC spots. Fisher's exact test was applied independently to each sample using `fisher.test` function in R, with a p-value < 0.05 considering as significant. Assuming the that the entropy distribution and GSC distribution are independent, the expect overalp between high entropy and high GSC spots were calculated as follows.

$$\text{Expected value} = N \times 0.25 \times 0.25 \quad (2.5)$$

where N represents the total number of spots per sample. 0.25 corresponds to the proportion of spots exceeding the 75th percentile threshold for each feature independently.

Observed overlap values were calculated as the number of spots simultaneously classified as high entropy and high GSC spots. Ratio between the observed and expected overlaps were computed to evaluate whether high entropy and high GSC regions co-occured more or less frequently than expected by chance.

2.7 Visualization of Spatial Transcriptomic Data

Spatial visualization of discrete entropy regions were performed using `SpatialDimPlot` function from Seurat [25] package. Entropy groups were visualized using a colour gradient ranging from low entropy regions to high entropy regions. `SpatialFeaturePlot` from the same package was used to plot mean trasncriptional entropy distributions, GSC module scores and GBM subtype associated module scores. Both functions are wrapper functions for `SpatialPlot` function which plots a feature or discrete cluster assignment for each spot over the tissue image.

3. Results

3.1 Distribution of Entropy across GBM samples

3.1.1 Mean entropy distribution

Figure 3.1 shows the distribution of mean transcriptional entropy across all samples. GBM6 exhibited the highest median transcriptional entropy with a median of 0.001358, while GBM1 exhibited the lowest median entropy of 0.001245. Since GBM6 represents peritumoral tissue rather than tumor tissue, the elevated entropy observed is unlikely to reflect stemness associated transcriptional disorder. Instead, the increased entropy likely reflects the presence of different neural and glial transcriptional programs associated with astrocytes, oligodendrocytes and neurons within the peritumoral microenvironment. This supports the hypothesis that transcriptional entropy is strongly influenced by tissue context rather than universally representing stemness. Among the five tumor samples, GBM3 exhibited the highest median entropy followed by GBM5, GBM2, GBM4 and GBM1, suggesting that GBM3 possess the highest transcriptional complexity among the tumor samples while GBM1 have the lowest entropy. However, the differences in median entropy between tumor samples were relatively small, indicating that the global entropy levels remain broadly conserved across the GBM tumors.

GBM6 exhibited the highest IQR with a value of 1.76×10^{-4} , while GBM3 exhibited the lowest IQR of 5.32×10^{-5} . Among the tumor samples, GBM4 showed the largest IQR at 1.11×10^{-4} . IQR reflects the spread of entropy values across spots within each sample. Accordingly, GBM4 demonstrates the largest spot level entropy heterogeneity among tumor samples while GBM3 exhibiting a more narrower distribution. High IQR of GBM6 likely reflects the coexistence of multiple neural and glial transcriptional programs within the peritumoral tissue. Outlier spots with extremely low entropy values were identified in GBM1, GBM3 and GBM6. In tumor samples, these spots may represent highly transcriptionally ordered regions, while in GBM6 they may correspond to specialized neural cell populations within the peritumoral tissue.

The violin plot shown in figure 3.1 illustrates the distribution of the mean entropy across spots within each sample. GBM2, GBM4, GBM5 and GBM6 exhibited bimodal distribution characterized by two peaks, suggesting the presence of two transcriptionally distinct spot populations. In contrast, GBM1 and GBM3 exhibited a trimodal distribution with three peaks, indicating increased transcriptional complexity and the presence of multiple entropy associated spot populations. GBM6 exhibited a narrow distribution suggesting a more uniform entropy distribution consistent with the peritumoral tissue context. Most of the samples showed a separation into relatively high entropy and low entropy groups, suggesting the existence of spatially distinct states within the tissue.

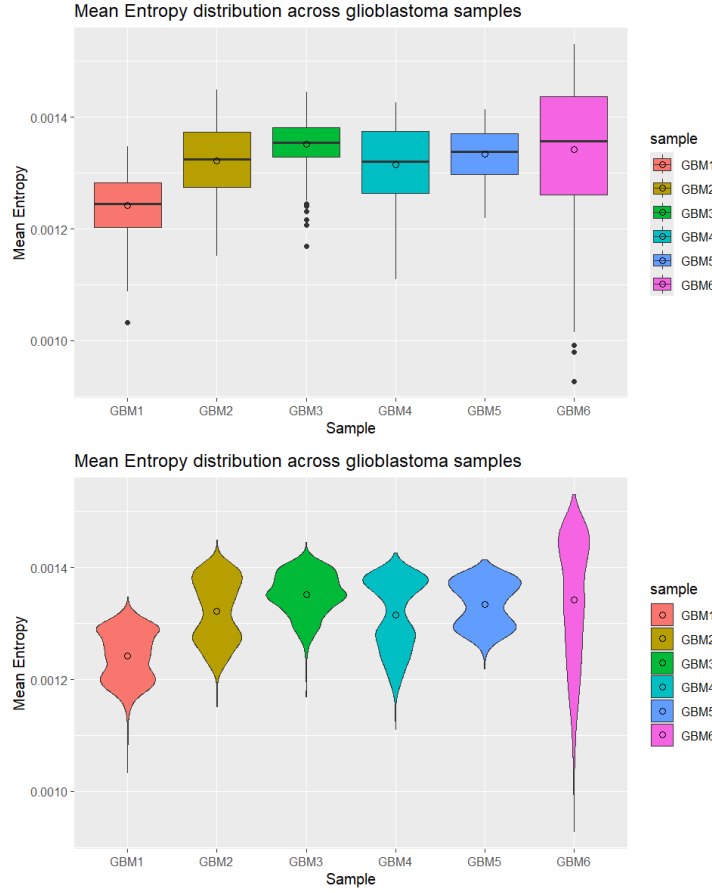


Figure 3.1: Distribution of mean transcriptional entropy across GBM samples. Top: Boxplot showing median entropy, IQR and outlier distribution across samples. Bottom: Violin plot showing density distribution of entropy values across spots within each sample.

3.1.2 Impact of Standard Deviation of Entropy Distribution

To further investigate the heterogeneity of entropy, standard deviation of entropy was incorporated with mean entropy to generate a combined score (Figure 3.2). Compared to the ranking based on mean entropy, GBM5 shifted to the lowest median, while GBM1 increased to the third highest ranking. The remaining samples retained similar relative rankings. This shift suggests that although GBM1 exhibited relatively low mean entropy, the presence of variably expressed genes increased the combined entropy score. In contrast, GBM5 showed both low entropy and low standard deviation entropy, resulting in a lower combined score. These findings indicate that incorporating the standard deviation of entropy captures additional transcriptional heterogeneity not identified using only mean entropy. Importantly, the incorporation of the standard deviation altered the distribution of outlier spots. In the mean entropy analysis, outliers were located within the low entropy region. However, they shifted towards the high entropy and high variability region when incorporated with the standard deviation of entropy. The consistent appearance of high entropy and high variability outlier spots suggests that they represent

biologically relevant transcriptionally heterogenous regions rather than technical artifacts.

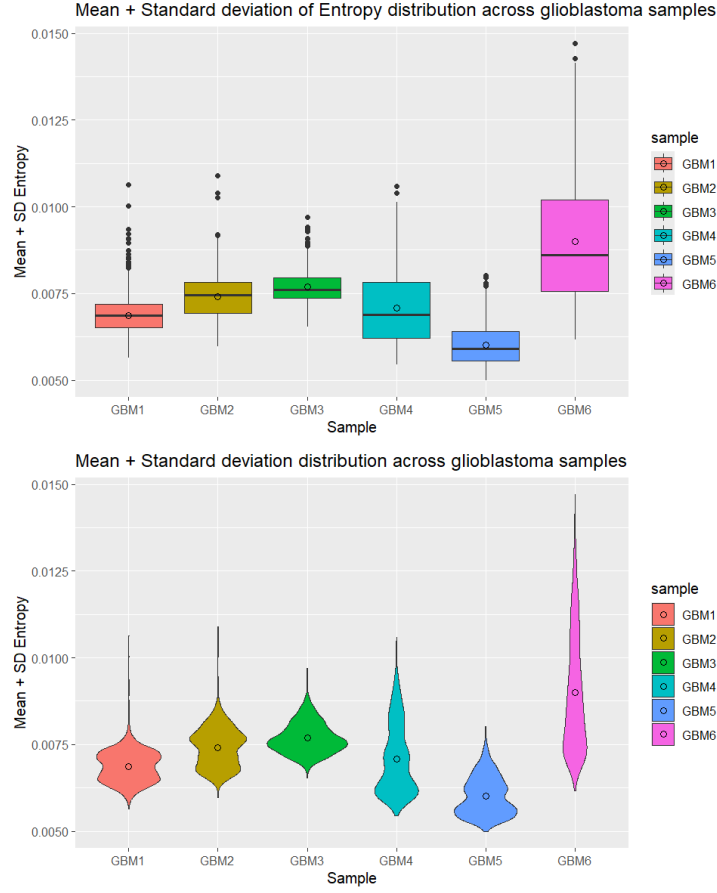


Figure 3.2: Distribution of combined entropy scores incorporating mean entropy and standard deviation across GBM samples. Top: Boxplot showing combined entropy score distributions and outlier profiles. Bottom: Violin plot showing density distributions of combined entropy scores across spatial spots.

The violin plot shown in (Figure 3.2) demonstrates multimodal distributions across all samples. GBM1, GBM2 and GBM3 exhibited bimodal distributions characterized by the presence of two peaks, suggesting the presence of two entropy populations. GBM4 and GBM5 exhibited trimodal distributions characterized by the presence of three peaks, suggesting the presence of three entropy based populations. In contrast, GBM6 exhibited a unimodal distribution, suggesting a more uniform entropy distribution consistent with the peritumoral tissue context. The multimodal distributions observed across the samples supports the use of GMM for entropy classification, as entropy could not be represented using a single Gaussian distribution.

3.1.3 Outlier detection and classification of transcriptional entropy

Outlier detection identified 18 outlier spots in GBM1, 5 in GBM2, 13 in GBM3, 2 in GBM4, 7 in GBM5 and 2 in GBM6. As described in Section 2.3, both mean entropy and standard deviation of entropy were incorporated into the GMM classification since the redistribution of

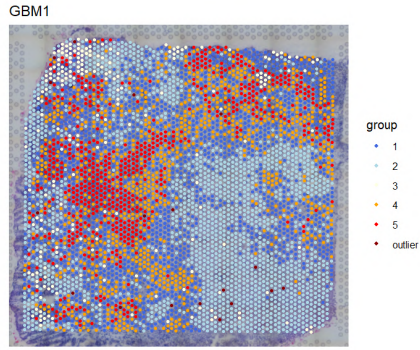
outliers observed after the incorporation of standard deviation suggested that standard deviation captured biologically relevant transcriptional heterogeneity without mean entropy alone. Following the removal of outlier spots, GMM classified the remaining spots into 3-8 groups per sample. Comparison of combined entropy scores of outliers with GMM component parameters showed that outlier spots exhibited entropy values exceeding the highest fitted component means. This suggests that the outlier spots represent high transcriptionally heterogeneous regions with elevated entropy and high gene level variability.

Spatial mapping of the groups revealed that transcriptional programs are not randomly distributed but instead formed coherent spatial domains across all tumor samples (Figure 3.3). This suggests that the entropy associated transcriptional states are organized by the TME and local cellular context. While all samples exhibited spatially organized domains, the number of groups classified varied across the samples, reflecting differences in transcriptional heterogeneity and complexity. GBM6 showed distinct spatial patterns consistent with peritumoral transcriptional programs, indicating region specific heterogeneity.

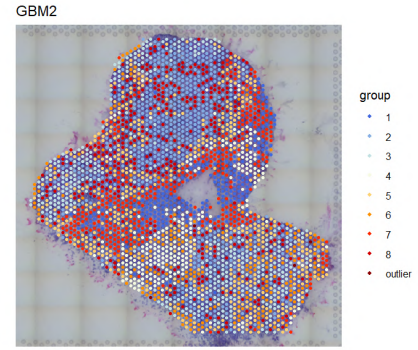
3.1.4 Functional enrichment analysis for discrete spatial entropy regions

Functional enrichment analysis was performed to characterize biological processes and pathways associated with discrete spatial entropy regions classified using GMM. Across all tumor samples, GO enrichment analysis enriched consistently for ribosomal and translational processes including cytoplasmic translation, ribosomal small subunit biogenesis and rRNA processing. These enrichments were highly significant across most entropy regions, suggesting a global transcriptional state across GBM. KEGG pathway enrichment further supported these findings by enriching for ribosome associated pathways. In addition to these pathways, several samples enriched for pathways annotated under viral infection, immune activation and inflammatory responses including Coronavirus disease, Salmonella infection, Type I diabetes mellitus and autoimmune thyroid disease pathways. These enrichments do not indicate the presence of the disease themselves but rather reflect the shared transcriptional programs associated with immune activation, inflammation and cellular stress response.

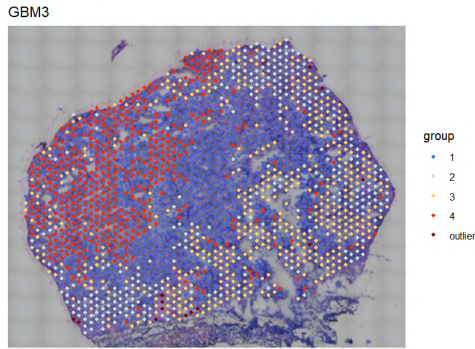
GBM1 (Figure 3.4) showed high and consistent GO enrichment for translational and ribosomal biological processes across all entropy groups, suggesting relatively conserved transcriptional programs despite spatial entropy differences. KEGG pathway enrichment showed consistent results along GO enrichment with additional enrichment of immune associated pathways in selected groups, indicating localized inflammatory transcriptional activity. GBM2 (Figure 3.5) demonstrated consistent GO enrichment for translational and ribosomal processes across all entropy regions. However, several entropy groups additionally enriched for immune associated pathways, suggesting increased immune related transcriptional heterogeneity within TME. GBM3 and GBM4 showed enrichment profiles similar to GBM1 with dominant enrichment of ribosomal and translation programs across entropy regions, suggesting conserved transcrip-



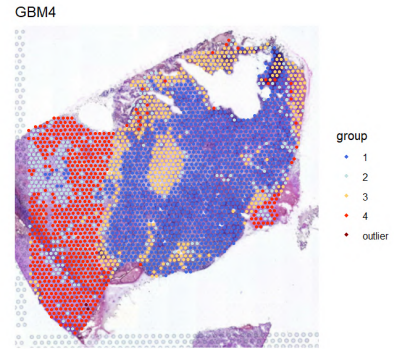
(a) GBM1



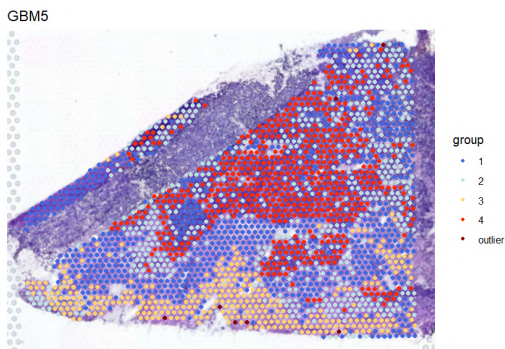
(b) GBM2



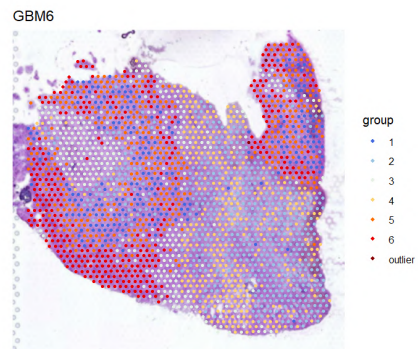
(c) GBM3



(d) GBM4



(e) GBM5



(f) GBM6

Figure 3.3: Spatial distribution of GMM-classified transcriptional entropy regions across GBM and peritumoral samples.

tional activity across tumor entropy states. GBM5 (Figure 3.6) resulted in a strong enrichment for ribosomal biogenesis and translational processes together with pathways associated with immune activation and cellular stress response. Some groups were enriched for neurodegeneration related pathways, suggesting having localized regions with elevated cellular stress, metabolic dysregulation or cell death associated transcriptional programs. GBM6 (Figure 3.7) demonstrated strong enrichment for ribosomal and translational activity. Unlike the tumor samples, several entropy regions enriched for pathways associated with neural function, neurodegeneration and stress response, likely reflecting the diverse neural and glial transcriptional programs associated with the peritumor tissue environment.

Overall, the enrichment analysis shows that ribosomal and translational programs represent a global transcriptional state across both tumoral and peritumoral entropy regions [41], whereas immune response, stress related and neurodegeneration pathways exhibit greater regional and sample specific variability [42]. These findings enabled further investigation of gene level entropy variability using standard deviation of entropy to identify transcriptional programs associated within each sample.

3.2 Gene level analysis using standard deviation of entropy

Genes exceeding the 90th percentile of standard deviation of entropy distribution at gene level were selected as high entropy variable genes. This resulted in 2090 genes in GBM1, 1949 in GBM2, 1922 in GBM3, 2028 in GBM4, 2094 in GBM5 and 1817 genes in GBM6. To focus on the most variable transcriptional states across each sample, the top 15 genes with highest entropy variability from each sample were selected for downstream comparative analysis.

3.2.1 Heatmap based clustering of entropy variable genes

To visualize shared and unique patterns of entropy variability, hierarchical clustering was performed on the top 15 variable genes and the results were displayed as a heatmap (Figure 3.8). Clustering identified three major gene clusters with distinct entropy variability patterns across the samples. Genes in the clusters were enriched using DAVID [28] and each cluster enriched for distinct transcriptional programs. First cluster enriched for functional profiles related to response to cellular stress, programmed cell death and immune signalling, with elevated levels primarily in GBM1 and GBM2. It also clustered mesenchymal markers *CHI3L1* and *VIM* [6, 7] suggesting that spatial variability of this cluster is driven by mesenchymal, immune and stress related transcriptional programs. Second cluster enriched for functional profiles related to neural cell proliferation and glial cell differentiation, with *GFAP* being a marker for astrocytes [43]. Spatial variability of these genes resulted moderate but elevated in GBM6 suggesting neural transcriptional programs being centred to GBM6 consistent with the peritumoral nature of GBM6. Third cluster enriched functional profiles related to cellular respiration and oxidative

phosphorylation.

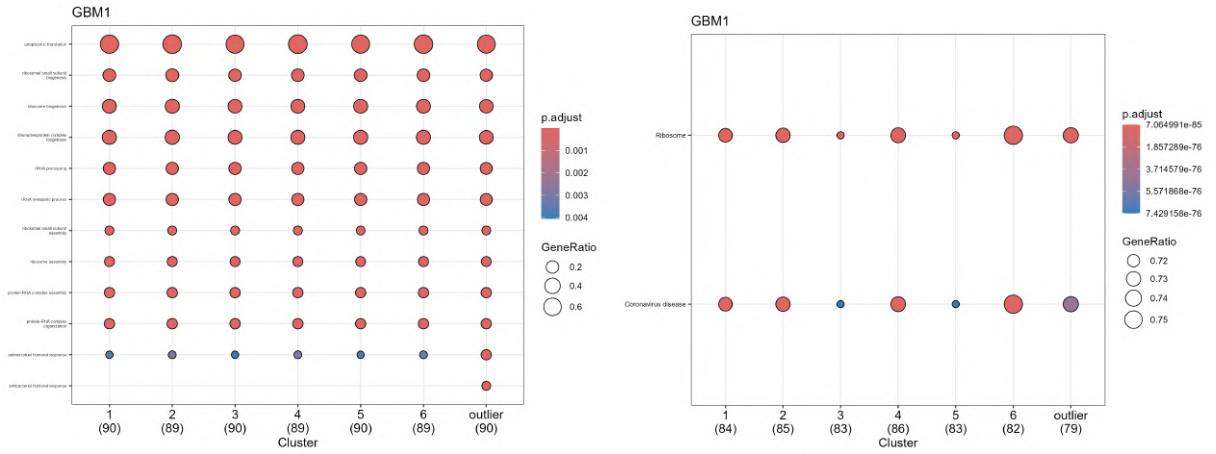


Figure 3.4: GO and KEGG pathway enrichment for GBM1

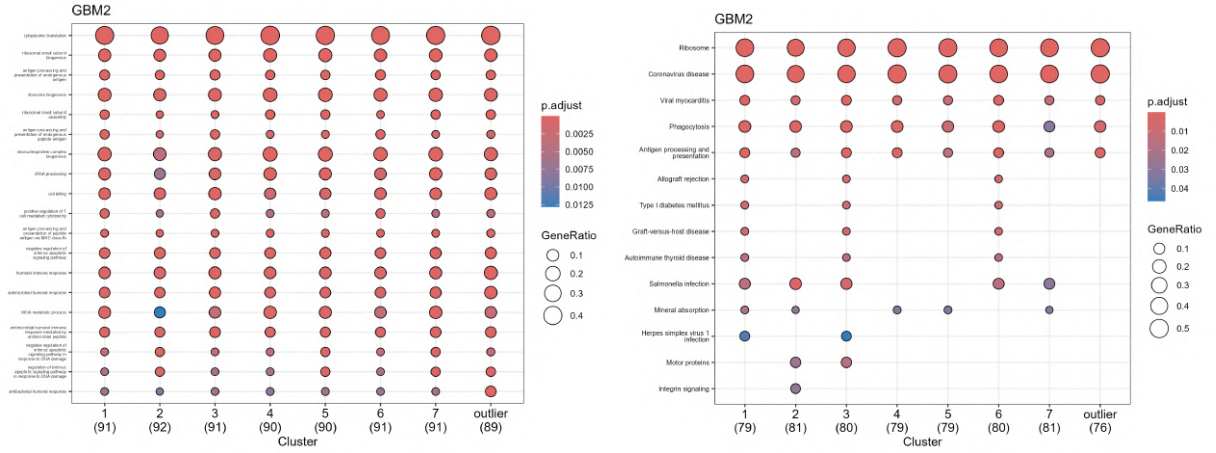


Figure 3.5: GO and KEGG pathway enrichment for GBM2

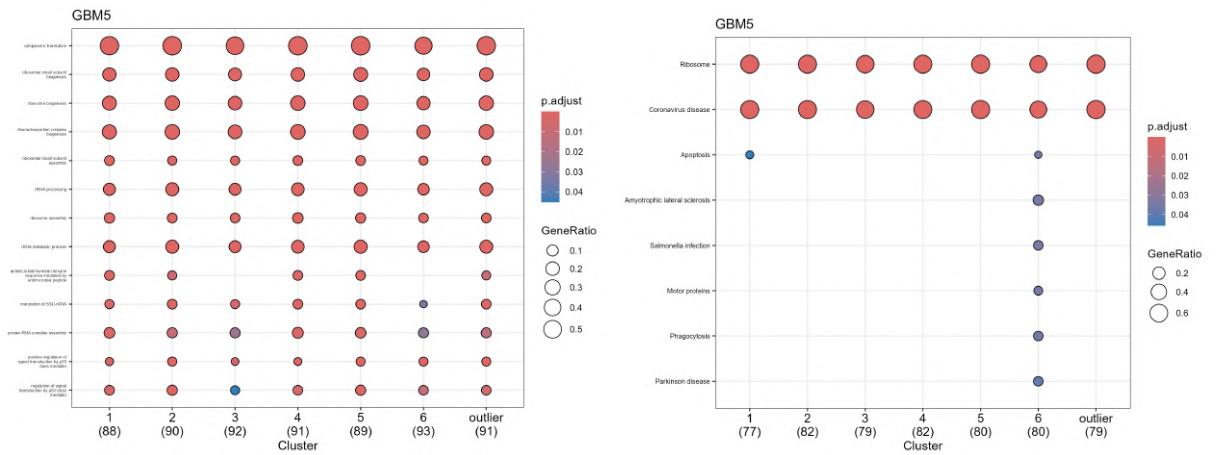


Figure 3.6: GO and KEGG pathway enrichment for GBM5

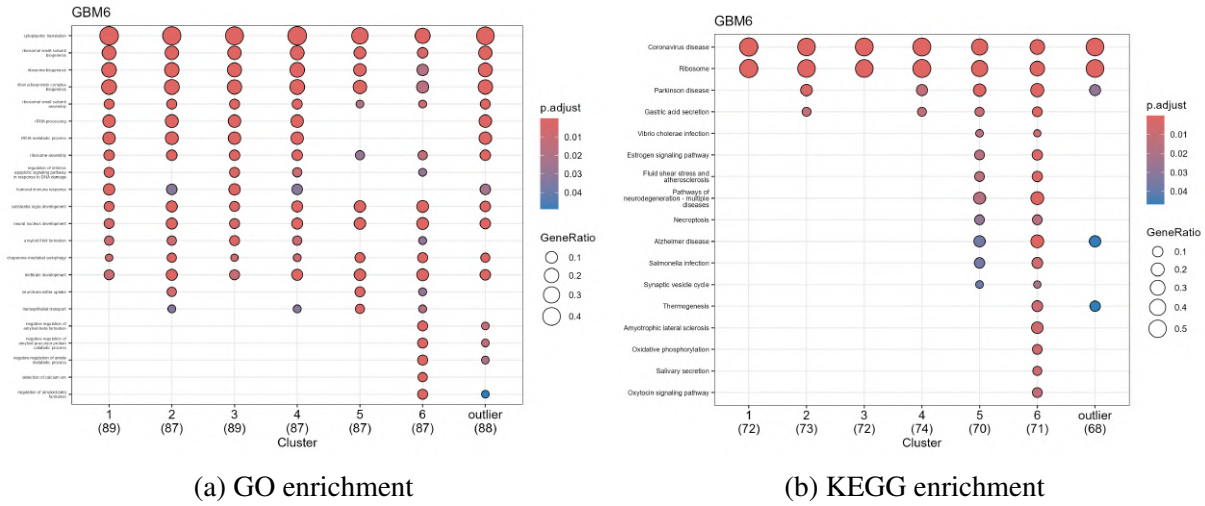


Figure 3.7: GO and KEGG pathway enrichment for GBM6

Standard deviation of Entropy per gene across all samples

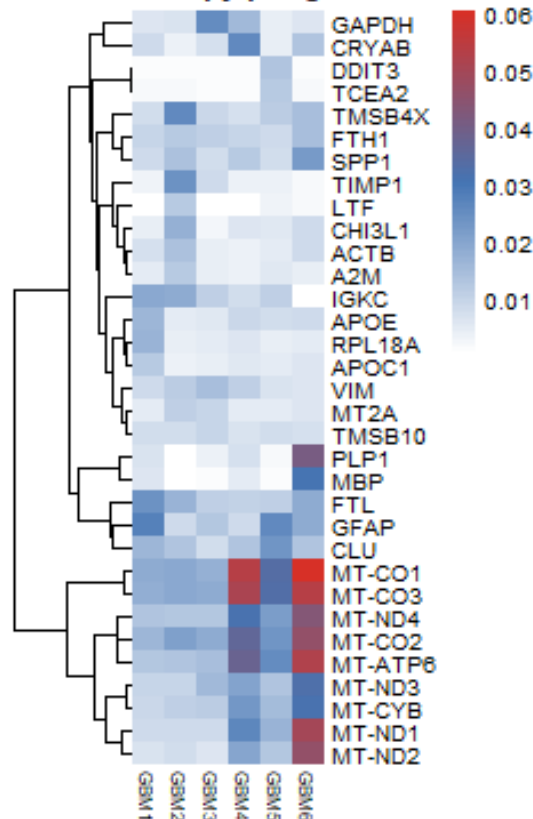


Figure 3.8: Heatmap of top 15 high standard deviation of entropy genes among all the samples

Cluster	Genes	Functional Profile
Cluster 1	<i>GAPDH, CRYAB, DDIT3, TCEA2, TMSB4X, FTH1, SPP1, TIMP1, LTF, CHI3L1, ACTB, A2M, IGKC, APOE, RPL18A, APOC1, VIM, MT2A, TMSB10</i>	Response to cellular stress, Programmed cell death, immune signalling
Cluster 2	<i>PLP1, MBP, FTL, GFAP, CLU</i>	neural cell proliferation, glial cell differentiation
Cluster 3	<i>MT-CO1, MT-CO3, MT-ND4, MT-CO2, MT-ATP6, MT-ND3, MT-CYB, MT-ND1, MT-MD2.</i>	Cellular respiration, oxidative phosphorylation

Table 3.1: Distinct high standard deviation entropy genes and their functional profile enriched using DAVID [28] in each cluster.

The overall results suggest that GBM6 spatial variability is mainly driven by neural development and metabolic related transcriptional programs while in tumor samples the spatial variability is mainly driven by immune, stress and inflammatory response. These findings indicate that entropy does not exclusively capture stemness but reflects context dependent biological programs influenced by tissue composition and microenvironment state.

3.2.2 Shared and sample specific entropy variable genes across samples

Genes that are shared across all the tumor samples were identified from high variable entropy genes beyond the 90th percentile. GO enrichment and KEGG pathway enrichment revealed strong enrichment for biological processes related to translational and cellular respiration, including aerobic respiration, oxidative phosphorylation, and ATP synthesis coupled electron transport (Figure 3.9). These results suggest that energy metabolism and protein synthesis represent conserved components of spatial transcriptional variability in glioblastoma. This observation is consistent with the recent study on disruption of glioblastoma metabolic pathways could result in impaired tumor progression by reducing cellular respiration [44]. This finding was further supported by KEGG pathway enrichment, which showed enrichment for ribosome and oxidative phosphorylation. In addition, pathways associated to neurodegenerative diseases such as Parkinson disease, Alzheimer disease and Huntington disease were also enriched, likely reflecting shared mitochondrial and metabolic pathways that also represent in neurodegenerative disease gene sets [45]. Overall, the results suggest that cellular respiration and neurodegeneration are conserved biological programs with glioblastoma spatial heterogeneity. A similar analysis was performed for genes shared across all samples, including peritumoral tissue, that exceeded the 90th percentile of high variable entropy genes. Both GO and KEGG enrichment revealed the same biological processes and metabolic pathways related to oxidative phosphorylation, ribosomal activity and cellular respiration (Figure 3.10), indicating that these biological

processes are preserved not only within tumor samples but also in peritumoral regions.

To identify sample specific transcriptional programs, genes that are unique to each sample were extracted from high entropy variable gene set exceeding the 90th percentile. Functional enrichment analysis (Figure 3.11) revealed distinct biological processes across most samples, indicating the contribution of different biological processes to spatial entropy variability in each tumor. While several samples showed enrichment for specific biological processes and pathways, no significant GO and KEGG enrichment was observed for GBM4 and no significant KEGG enrichment for GBM2 was identified. These findings suggest that both shared and sample specific biological programs contribute to transcriptional entropy heterogeneity in glioblastoma.

GBM1 was enriched for functional profiles related to DNA damage response, chromatin remodeling, and transcription regulation, suggesting that entropy variability in this sample is likely linked to mechanisms related to genome and epigenetic regulations. This indicates increased plasticity and genomic instability which are contributors to heterogeneity in glioblastoma [46]. These findings are consistent with classical subtype of GBM, described in the study of Verhaak et al. on glioblastoma subtypes [6]. This enrichment may reflect adaptive transcriptional programs that enable tumor cells to respond to DNA damage and microenvironmental stress by promoting survival under pressure. KEGG pathway was enriched for hepatocellular carcinoma and adherent junctions, suggesting overlap with oncogenic signaling and cell to cell adhesion disruption, which are features of malignant transformation and invasion [47].

GBM2 was enriched for processes related to endoplasmic reticulum (ER) to golgi vesicle mediated transport, lysosomal lumen function, negative regulation of substrate adhesion dependent signaling, focal adhesion and extracellular matrix (ECM) organization. These results suggests a strong functional role for intracellular trafficking and membrane remodelling that drives the entropy variability. Enrichment of adhesion related processes like focal adhesion and collagen containing ECM components suggests dynamic regulation of cell to matrix interactions which are essential for GBM invasiveness and associated with GBM infiltrative behaviour and therapy resistance [48].

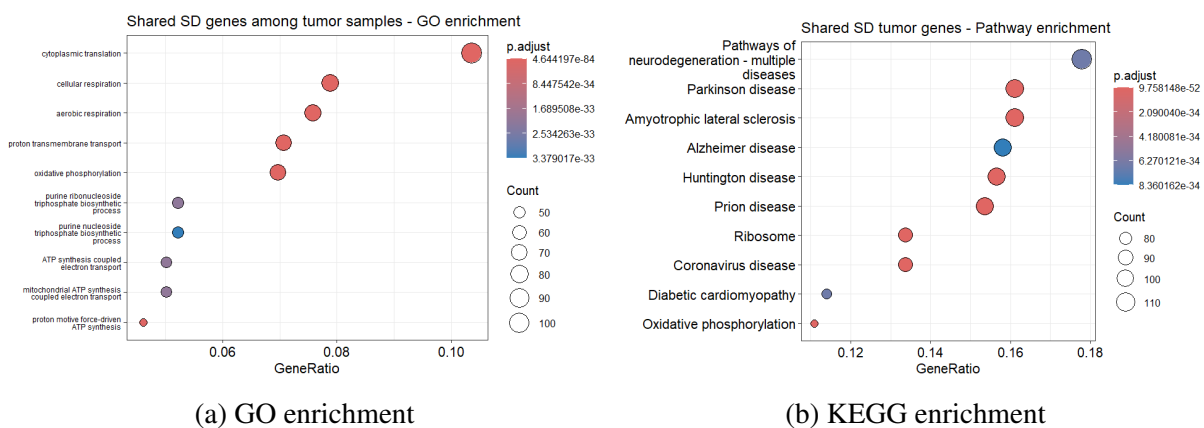


Figure 3.9: GO and KEGG pathway enrichment for genes shared across tumor samples

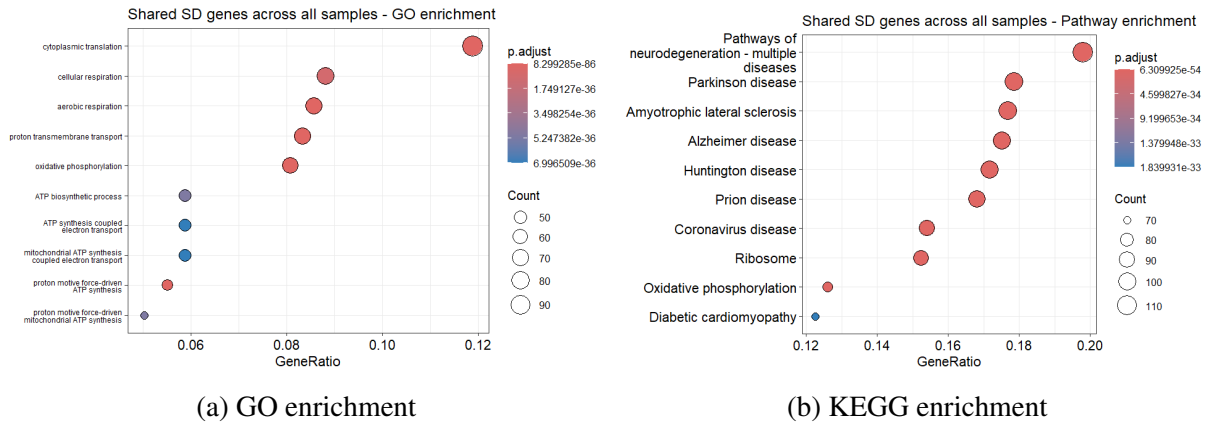
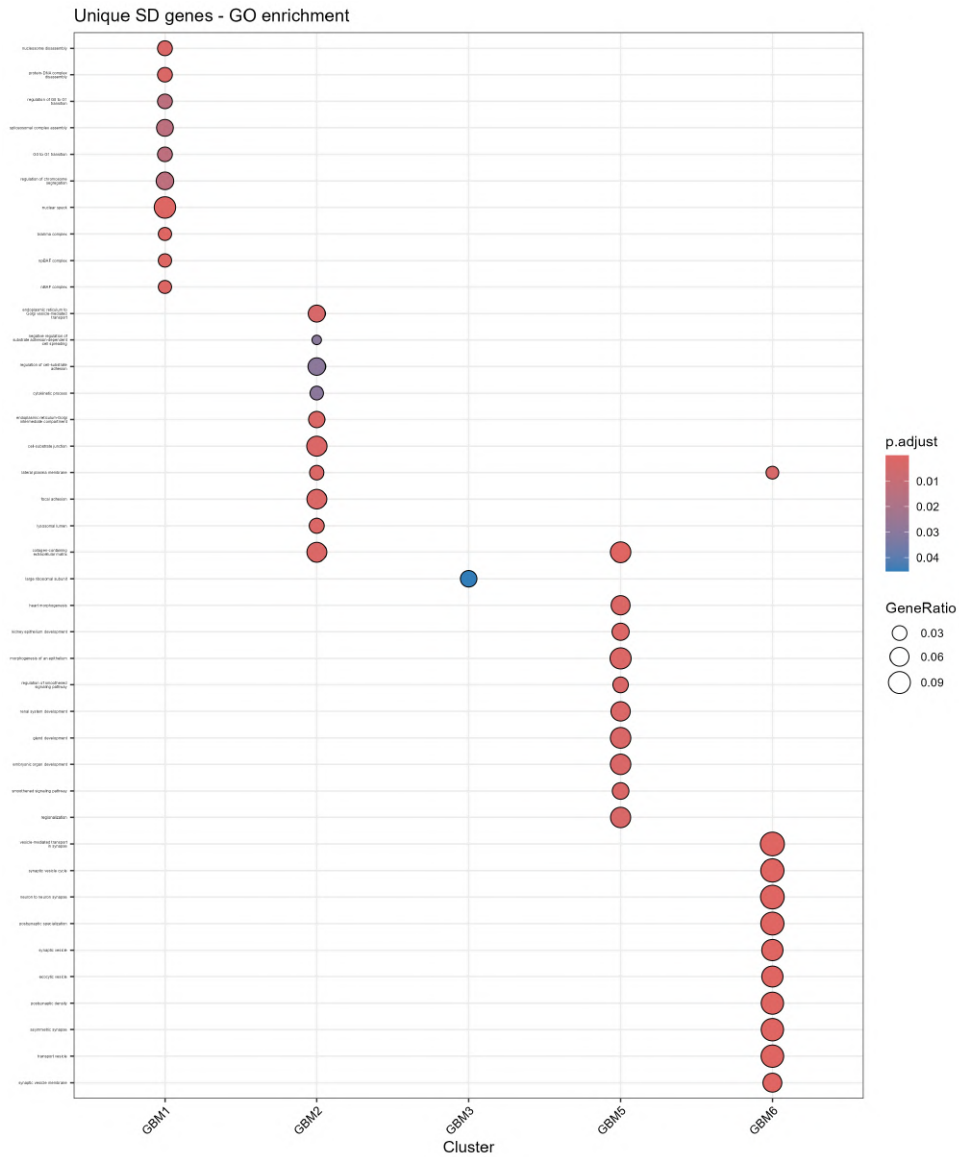


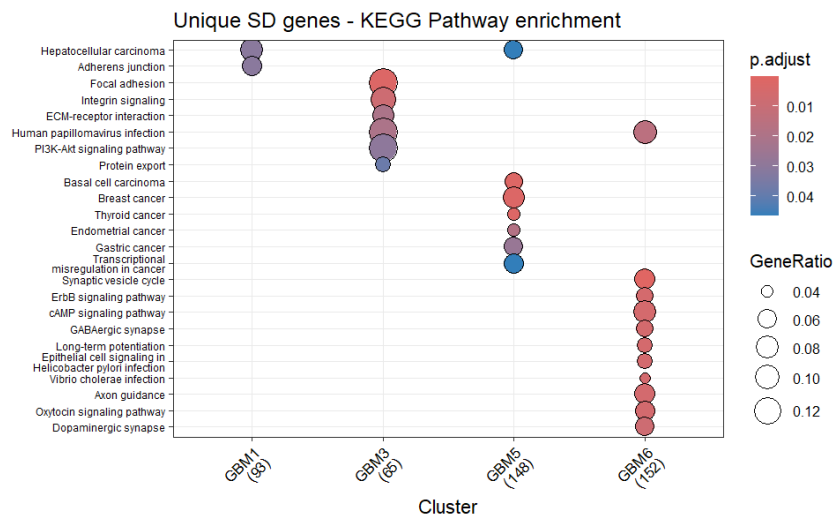
Figure 3.10: GO and KEGG pathway enrichment for genes shared across all the samples

GBM3 was enriched for large ribosomal subunit, indicating high translational machinery activity. This suggests elevated protein synthesis, which is commonly associated with proliferative tumor states. KEGG pathway enrichment resulted focal adhesion, integrin signaling, ECM receptor interaction, human papilloma virus infection and PI3K-AKT signaling pathway. These pathways indicate strong activation of growth factor signaling, survival pathways and ECM interactions. PI3K-AKT pathway is one of the most frequently activated signaling cascades in GBM and plays a crucial role in promoting proliferation, survival and metabolic adaptation [49]. Integrin and ECM interaction also critical mediators of tumor invasion and TME communication [50]. Enrichment of protein export pathways suggests increased secretory or membrane trafficking activity supporting tumor progression and intercellular communication [49].

GBM4 did not result statistically significant enrichment in neither GO nor KEGG pathway enrichments, suggesting a more heterogeneous or transcriptionally neutral entropy state in the sample. It could also reflect a distributed set of weak signals that do not cluster dominant biological pathways. Such result could be observed sometimes in highly stable or less transcriptionally active regions [51]. GBM5 was highly enriched for development and morphogenic processes such as collagen containing ECM organization, kidney and gland development, epithelial morphogenesis and embryonic organ development. Additionally, regulation of smoothened signalling pathway and regionalization processes were significantly enriched. These results suggest the reactivation of developmental signalling programs associated with embryogenesis and tissue patterning. Smoothen pathway is a key component of the Hedgehog signalling which shows stem-like properties and tumor progression in glioblastoma [52]. KEGG pathway enrichment resulted in multiple cancer related pathways like hepatocellular carcinoma, basal cell carcinoma, breast cancer, thyroid cancer, endometrial cancer, gastric cancer and transcriptional misregulation in cancer. This suggests the presence of shared oncogenic signaling pathways across the tumor indicating that GBM5 may represent a highly plastic tumor state. Developmental pathway activation and ECM remodeling are well established features of GSC and are associated with tumor aggressiveness and therapeutic resistance [52].



(a) GO enrichment



(b) KEGG enrichment

Figure 3.11: GO and KEGG functional enrichment for genes that are unique to each sample

GBM6 was enriched for synaptic and neuronal related processes such as vesical mediated transport in synapses, synaptic vesicle cycle, neuron to neuron synapses, postsynaptic specialization, synaptic vesicle, post synaptic density and axon guidance. This enrichment indicates strong neural like features within the peritumor [53]. KEGG pathway enrichment further supports the GO enrichment results, enriching for synaptic vesicle cycle, ErbB signalling pathway, cAMP signalling pathways, GABergic synapses and potentiation. GBM6 highlights the peritumoral microenvironment characterized by increased synaptic signaling and neural interaction [54].

3.3 Spatial distribution of GBM biological signatures

To investigate whether transcriptional entropy was associated with known glioblastoma transcriptional programs, module scores were calculated for GSC, mesenchymal, proneural, classical and neural signatures. Spatial distributions of these module scores were compared with mean transcriptional entropy spatial plots to identify regions of spatial overlap (Figure 3.12 - 3.14). In GBM1 (Figure 3.12a), the spatial distribution of GSC module score appeared heterogeneous with a partial overlap with high transcriptional entropy regions. In contrast, partial overlap was observed between mesenchymal module scores and high entropy regions. Mesenchymal signatures were concentrated towards the lower central region of the tissue, suggesting localized activation of aggressive transcriptional programs [6, 7]. Proneural signatures showed partial spatial overlap with high entropy regions and were positioned adjacent to mesenchymal region. The close spatial relationship between proneural and mesenchymal signatures may represent a proneural-mesenchymal transition (PMT) like spatial organization [7]. Classical signature was distributed relatively uniformly, while neural signatures were distributed with relatively low module scores.

In GBM2 (Figure 3.12b), no overlap was observed between high entropy regions and high GSC module scores. High transcriptional entropy regions were observed to be clustered to the upper left and the lower right portions of the tissue. Considering the enrichment of immune related biological processes among sample specific genes, these entropy rich regions may reflect immune associated transcriptional activity rather than stemness associated heterogeneity. Both GSC and mesenchymal signatures were distributed heterogeneously without distinct spatial clustering. Although, limited overlap was observed between mesenchymal signatures and high entropy regions, areas with high GSC scores appeared to occur within low entropy regions. Proneural and neural signatures showed relatively very low module scores throughout the tissue, while classical signatures showed a relatively uniform and moderate module score distribution. Overall, these findings suggest that entropy in GBM2 is more strongly associated with immune related microenvironmental processes rather than GSC associated transcriptional programs, consistent with a mesenchymal like TME [4].

In GBM3 (Figure 3.13a), a significant overlap was not observed between high transcrip-

tional entropy regions and high GSC module scores. However, mesenchymal and proneural signatures showed adjacent and partially overlapping spatial distributions similar to those observed in GBM1. These regions coincided with high transcriptional entropy regions, suggesting spatial coexistence of proneural and mesenchymal transcriptional programs. Mesenchymal signatures appeared more prominent towards the tissue periphery and gradually decreased towards the core likely reflecting spatial variation in invasiveness [48, 50]. Both classical and neural signatures were uniformly distributed with low module score. Overall, the spatial organization suggests an intermediate state characterized by coexistence of proneural and mesenchymal transcriptional programs [7].

In GBM4 (Figure 3.13b), spatial overlap was observed between high transcriptional entropy regions and high GSC module scores. High entropy regions also coincided with regions with high proneural module scores, suggesting that entropy is associated with stemness and active transcriptional reprogramming. Spatial comparison of mesenchymal and proneural signatures revealed a striking organization in which high mesenchymal module score regions formed a central core surrounded by high proneural module score regions (Figure 3.15a). The interface between these two transcriptional states corresponded closely with regions of high transcriptional entropy and high GSC module scores. This pattern is consistent with an active PMT region, where cells may be transitioning between proneural and mesenchymal states [7]. Spatial overlap of both high entropy and high GSC module scores at this boundary suggests that transcriptional entropy captures biologically dynamic regions characterized by cellular plasticity and subtype transition. Classical signatures remained uniform with low module scores, whereas neural signatures were nearly absent. These findings support the hypothesis that transcriptional entropy may identify spatially localized regions of the tumor that are associated with stem-like cells and subtype transitions [1, 37].

In GBM5 (Figure 3.13c), an inverse relationship was observed between entropy and GSC module scores. Regions with elevated entropy corresponded to regions with low GSC module scores and vice versa. Furthermore, high GSC module score regions overlapped with high mesenchymal module score regions. A small cluster of high proneural module score regions overlapped with high mesenchymal module score regions while most proneural signatures remained spatially distinct (Figure 3.15b). Classical signatures showed moderate distribution, while neural signatures remained uniformly low. In GBM6 (Figure 3.14), no meaningful overlap was observed between transcriptional entropy and either GSC and glioblastoma subtypes. Most signatures showed uniform distributions with low to moderate module scores. However, neural signatures resulted two distinct high neural module scores regions that partially overlapped with high transcriptional entropy regions (Figure 3.16). Based on the functional enrichment results for this sample and patterns of spatial distribution overlaps, it suggests that the entropy within this sample is primarily driven by neural transcriptional programs rather than stemness associated processes, consistent with the biological role of GBM6 as peritumoral sample [23].

Overall, the relationship between transcriptional entropy and GBM transcriptional programs

varied substantially across samples, reflecting underlying biological heterogeneity. Samples with PMT like organization, specially GBM4, demonstrated high transcriptional entropy at PMT interface together with partial overlap with high GSC module score regions, suggesting that entropy captures transcriptional instability associated with cellular plasticity and stemness. In contrast, high transcriptional entropy regions in immune associated samples such as GBM2 appeared largely independent of GSC activity and were more likely influenced by immune driven TME. Similarly, the inverse relationship observed in GBM5 indicates that high transcriptional entropy does not universally correspond to stemness associated programs. Overall, these findings support the hypothesis that transcriptional entropy is highly context-dependent and reflect distinct biological processes depending on tumor subtype composition and local tissue environment.

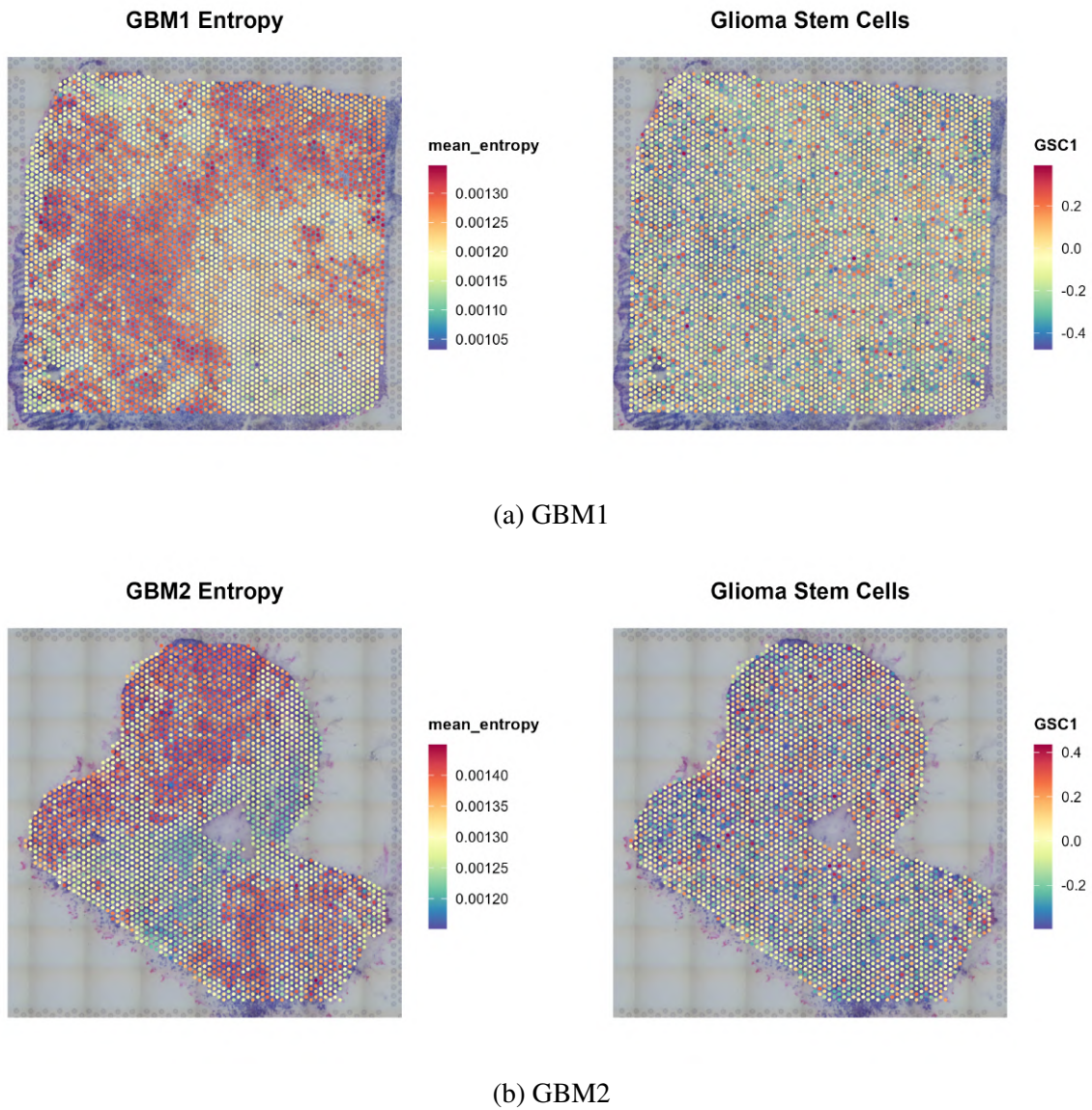
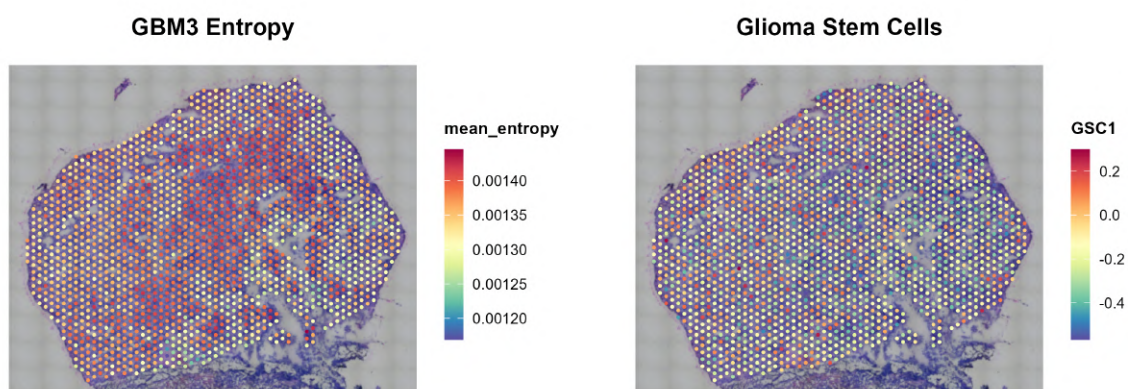
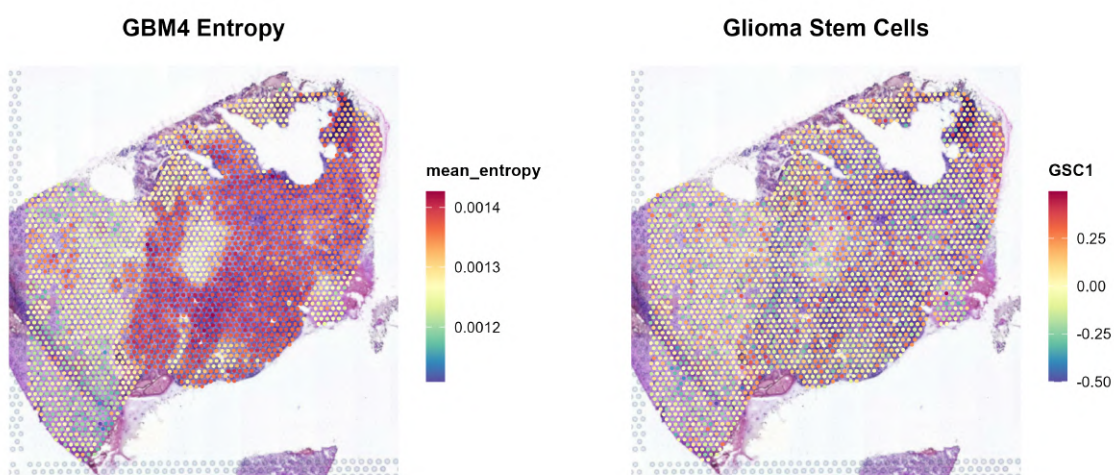


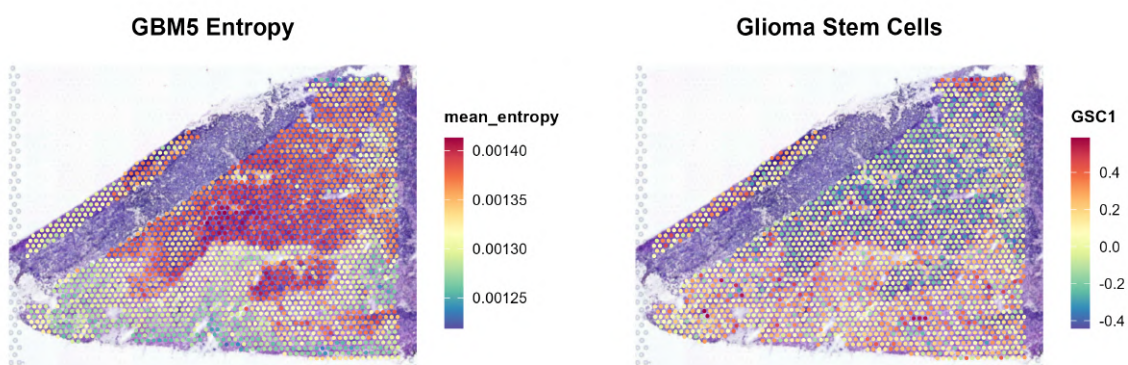
Figure 3.12: Module score distribution of Transcriptional Entropy and GSC (GBM1 and GBM2)



(a) GBM3



(b) GBM4



(c) GBM5

Figure 3.13: Module score distribution of Transcriptional Entropy and GSC (GBM3 - GBM5)

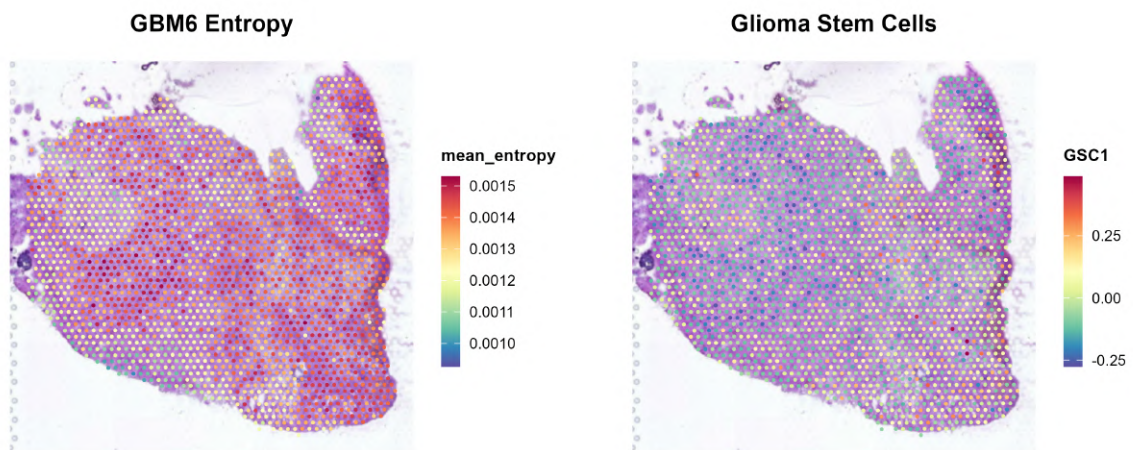
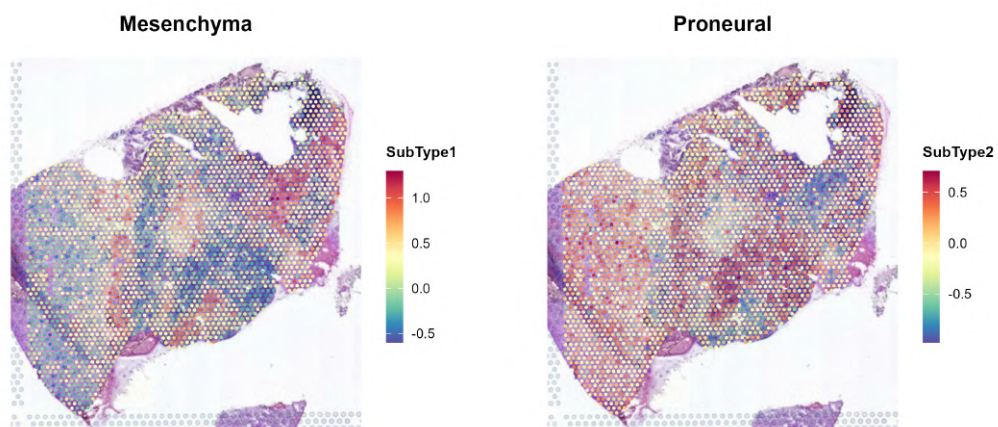
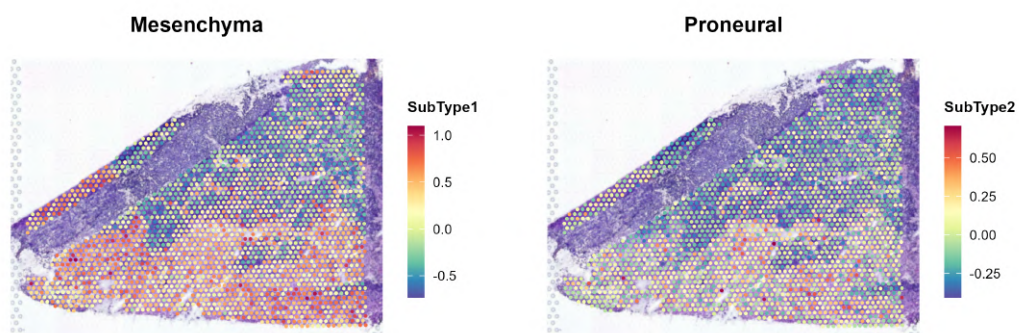


Figure 3.14: Module score distribution of Transcriptional Entropy and GSC for GBM6



(a) GBM4



(b) GBM5

Figure 3.15: Module score distribution of mesenchymal and proneural signatures for GBM4 and GBM5

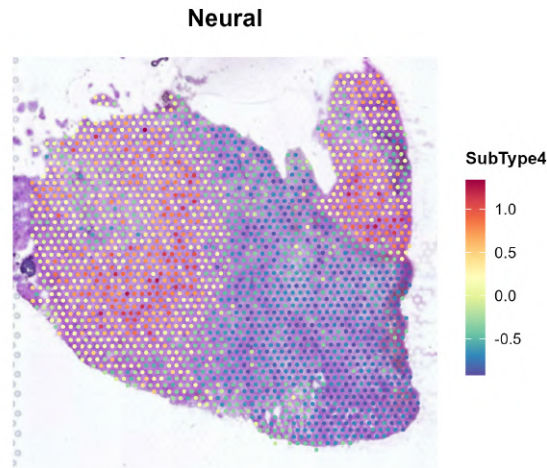


Figure 3.16: Spatial distribution for neural signature in GBM6

Spatial comparison between discrete entropy regions and transcriptional entropy regions demonstrated that clustering approach successfully captured the major entropy gradients observed within each sample. High entropy groups consistently localized into regions with high transcriptional entropy, while low entropy groups correspond to regions with low entropy values.

3.3.1 Correlation between entropy and GBM biological signatures

To quantify the relationship between transcriptional entropy and GBM biological signatures, Pearson correlation coefficients were calculated between mean transcriptional entropy and module scores for GSC, mesenchymal, proneural, classical and neural signatures using `cor()` function in R. Correlation coefficients range from -1 to 1, where positive values indicate that high entropy regions tend to have high module scores while negative values indicate that high entropy regions tend to have low module scores. The results are summarized in Table 3.2.

GBM1 showed negligible correlation between entropy and GSC, mesenchymal and neural signatures, while weak positive correlations were observed in proneural and classical signatures. These findings suggest a slight tendency for high entropy regions to coincide with proneural and classical transcriptional programs. However, the overall weak correlations indicate that the transcriptional entropy is largely independent of subtype specific and stemness associated signatures in this sample. This is consistent with spatial distribution, where only partial overlap between entropy and subtype specific associated regions were observed.

GBM2 showed negligible correlations among all signatures indicating that transcriptional entropy is largely independent of GBM subtypes and stemness associated transcriptional programs. This is consistent with spatial distributions, where high entropy regions appeared largely distinct from all biological signatures. Together with the functional enrichment results, this

suggests that entropy variability in GBM2 may be influenced more strongly by TME than by subtype specific transcriptional programs.

GBM3 exhibited a mixture of weak positive and weak negative correlations. Entropy remained independent of GSC activity but showed weak positive associations with proneural and classical signatures, while weak negative associations were observed with mesenchymal and neural signatures. These findings suggest a modest tendency for high entropy regions to coincide with proneural associated transcriptional activity. This is consistent with spatial distributions, where partial overlap between entropy and proneural regions was observed.

GBM4 showed the strongest positive association between entropy and GSC scores among all tumor samples. Correlation with mesenchymal and proneural signatures remained negligible, while a weak positive association was observed with classical signatures. In contrast, a strong negative correlation was observed with neural signatures. The positive association between these features suggest that high entropy regions may be associated with stem like transcriptional activity. The negligible correlations with mesenchymal and proneural signatures further suggest that transcriptional entropy may be associated with the transition interface between these transcriptional states rather than either subtype individually.

GBM5 exhibited negative correlations across all signatures. In particular, moderate negative associations were observed between entropy and GSC, mesenchymal and neural signatures. These findings indicate that high entropy regions are largely distinct from GBM subtype and stemness associated transcriptional programs. This pattern is consistent with the spatial distributions where regions enriched for GSC and mesenchymal signatures tended to occur in areas with lower entropy. The inverse relationship observed in GBM5 contrasts with the positive association between entropy and GSC activity observed in GBM4.

GBM6 showed near zero correlations across all signatures, indicating that transcriptional entropy is largely independent of both stemness associated and subtype associated transcriptional programs within the peritumoral tissue. This is consistent with the spatial distributions and suggests that entropy in GBM6 reflects biological variation distinct from transcriptional programs captured by GBM subtype classifications.

Overall, the correlation analysis demonstrated that the relationship between transcriptional entropy and GBM biological signatures is highly context-dependent. Most samples, showed weak or negligible correlation indicating that entropy captures biological variation largely independent of established GBM transcriptional programs. GBM4 was the only tumor sample showing a notable positive association between entropy and GSC activity, whereas GBM5 exhibited moderate negative correlations across multiple signatures. Together, these findings suggest that transcriptional entropy does not universally represent stemness associated transcriptional activity but distinct biological processes depending on the tumor context.

Sample	GSC	Mesenchymal	Proneural	Classical	Neural
GBM1	0.031	0.023	0.166	0.117	0.005
GBM2	- 0.038	- 0.040	- 0.022	0.026	- 0.040
GBM3	0.000	- 0.133	0.154	0.132	- 0.119
GBM4	0.228	0.044	0.026	0.101	- 0.647
GBM5	- 0.380	- 0.502	- 0.297	- 0.166	- 0.567
GBM6	0.019	0.091	- 0.025	- 0.128	- 0.040

Table 3.2: Correlation between transcriptional entropy and GSC and glioblastoma subtypes

3.3.2 Fisher’s exact test of transcriptional entropy and GSC spatial overlap.

To determine whether observed overlap between high mean entropy and high GSC regions occurred more frequently than expected by chance, Fisher’s exact test [39] was performed for each sample. The observed number of overlapping spots was compared to the expected number of overlaps assuming a random spatial distribution. The results are summarized in Table 3.3

GBM1 did not show a significant overlap between high mean entropy and high GSC regions. Although slight overlap was observed in the spatial plots, the overlap was not greater than expected by chance. This confirms that entropy and GSC expression are largely independent in this sample.

GBM2 also did not show a significant overlap. This is consistent with the spatial distributions, where high entropy regions appeared distinct from high GSC module score regions. Together with the correlation analysis and functional enrichment results, this suggests that entropy in GBM2 is largely independent of stemness associated transcriptional programs.

GBM3 did not show a significant overlap between entropy and GSC regions. The observed overlap was slightly lower than expected by chance (Observed/ Expected = 0.89). This is consistent with the correlation analysis, which showed that entropy remained independent of GSC activity. Although spatial organization suggested a possible PMT like architecture, statistical analysis indicated that entropy and stemness are not significantly associated with this sample.

GBM4 showed a significant overlap between high mean entropy and high GSC regions. The observed overlap was approximately 31% higher than expected by chance (Observed/ Expected = 1.31). This finding confirms that regions with high mean entropy and high GSC activity co-occur more frequently than expected under a random distribution. Combined with spatial organization of proneural and mesenchymal regions, these results support the presence of a biologically active transition zone enriched for stem-like cells.

GBM5 showed highly significant overlap between high mean entropy and high GSC module scores regions. The observed overlap was approximately 61% lower than expected by chance (Observed/ Expected = 0.3995). This confirms the strong spatial segregation between entropy rich and GSC enriched regions observed in the spatial plots. Unlike GBM4 where transcriptional entropy and stemness co-localized, GBM5 exhibited a clear separation between these

features, suggesting a different relationship between entropy and stemness within this tumor.

GBM6 showed a modest but statistically significant overlap between mean entropy and GSC regions. The observed overlap was approximately 16% higher than expected by chance (Observed/ Expected = 1.157). This finding was unexpected given the near zero correlation between entropy and GSC scores. The result suggests that although entropy and GSC expression are not globally associated across the tissue, localized regions of overlap occur more frequently than expected by chance.

Overall, Fisher’s exact test [39] demonstrated that the relationship between transcriptional entropy and stemness is highly context dependent. While GBM4 showed significant overlap between high mean entropy and high GSC module score regions, GBM5 showed significant spatial segregation between these features. The remaining samples showed either no significant association or only modest overlap. Together, these findings demonstrate that the same entropy metric can be associated with distinct biological states depending on the tumor subtype composition and local tissue context.

Sample	Observed	Expected	Ratio	P-value
GBM1	316	297.75	1.061	1.64×10^{-1}
GBM2	185	180.69	1.024	6.92×10^{-1}
GBM3	99	110.94	0.892	1.29×10^{-1}
GBM4	208	158.75	1.310	3.48×10^{-7}
GBM5	55	137.69	0.3995	1.14×10^{-23}
GBM6	157	135.69	1.157	1.63×10^{-2}

Table 3.3: Fisher’s exact test results between high transcriptional entropy and high GSC region overlaps

4. Discussion

This study investigated spatial transcriptional heterogeneity in glioblastoma using transcriptional entropy and compared with known GBM biological signatures. Overall, the results demonstrate that GBM is highly heterogeneous both spatially and biologically, with the interpretation of transcriptional entropy varying between samples. A key finding of this study is that transcriptional entropy does not represent a single biological process such as stemness but instead, it depends on the local tumor context. While GBM4 showed significant spatial overlap between high transcriptional entropy and high GSC regions, specially at PMT interfaces, GBM5 showed the opposite pattern with strong spatial segregation between transcriptional entropy and GSC activity. These findings indicate that entropy may reflect stemness and cellular plasticity in some tumors but different biological processes in others.

The remaining tumor samples further support this observation where GBM1, GBM2 and GBM3 showed weak or negligible associations between entropy and GBM signatures. In particular, entropy in GBM2 appeared to be more associated to immune related TME activity rather than tumor stemness. In contrast, GBM6 which was identified as a peritumoral tissue showed distinct patterns where entropy was mainly associated with neural and synaptic related processes, suggesting that entropy in non tumor tissue reflects neural transcription diversity than tumoral heterogeneity.

Gene level analysis revealed both shared and sample specific biological programs. Across all samples, ribosomal and translational processes were consistently enriched in entropy variable regions. This suggests shared transcriptional programs across both tumoral and peritumoral tissue. However, other processes such as immune signaling, metabolic activity and developmental pathways were highly variable between samples, highlighting functional heterogeneity. Overall, the results show that transcriptional entropy is not a direct marker of stemness but instead reflects multiple biological processes depending on spatial context.

Limitations of this study include the small sample size, which limits generalization and spatial overlap analysis for a limited number of GBM biological signatures. In addition, spatial resolution limits prevent true single cell resolution and pathway analysis depends on existing annotations which may miss novel mechanisms. The study also shows associations only and does not prove causality. Future work should include large sample size, integration with scRNA-seq data and comparative analysis with other GBM datasets. Multi-omics integration and machine learning approaches could further improve interpretation and help link entropy patterns to clinical outcomes. Overall, this study shows that transcriptional entropy captures biologically meaningful spatial heterogeneity in GBM. However, its biological interpretation depends strongly on tumor context and would be considered together with spatial, functional and molecular information.

5. Conclusion

This study used spatial transcriptomic data to analyze transcriptional heterogeneity in GBM using transcriptional entropy combined with functional and signature based analyses. The results show that transcriptional entropy captures spatial variation across GBM samples, reflecting multiple biological processes rather than a single underlying mechanism.

The relationship between transcriptional entropy and stemness varied between samples. GBM4 showed strong spatial overlap between high transcriptional entropy and GSC enriched regions, with evidences associated to PMT states, suggesting that entropy can identify biologically dynamic regions associated with cellular plasticity. In contrast GBM5 showed strong spatial segregation between entropy and GSC, suggesting that high transcriptional entropy does not universally correspond to stemness. Other samples showed weak or negligible associations between transcriptional entropy and GBM subtypes, highlighting the context dependent nature of transcriptional entropy.

Across all samples, ribosomal and translational pathways were consistently enriched, indicating shared transcriptional programs in both tumor and peritumoral tissue. GBM6 showed distinct patterns driven by neural and synaptic processes, confirming the strong influence of tissue type on entropy. Overall, transcriptional entropy is not a universal marker of stemness but a context dependent measure of transcriptional variability influenced by tumor state and TME.

In conclusion, this study indicates that transcriptional entropy is not a universal marker for GBM stemness but rather a context-dependent measure of transcriptional variability influenced by tumor subtype, cellular plasticity and TME. Transcriptional entropy provides a useful framework when integrated with spatial and functional analyses for investigating complex biological heterogeneity that characterizes GBM .

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